Enhancing Longitudinal Medical Image Segmentation through Spatial-temporal Fusion

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***Abstract*—Medical imaging research has seen advances in deep learning, but temporal aspects in time-series medical imaging data are often overlooked, leading to diagnostic limitations. This study proposes a spatial-temporal fusion approach for medical image segmentation by integrating a 3D UNet spatial network with a novel temporal network, DTransformer, capable of handling irregularly spaced sequences. The 3D UNet extracts spatial features, while DTransformer processes temporal information with time distance considerations using a novel self-attention mechanism. Experiments on a lung CT dataset show significant segmentation accuracy improvements with the fusion approach. DTransformer proves effective for unequally spaced sequences and boosts performance. And spatial-temporal fusion enhances medical image segmentation. Moveover, DTransformer's ability to manage temporal context and time distance holds promise for various tasks, indicating a new avenue for research.**

***Index Terms*—Transformer, temporal analysis, spatial-temporal fusion, time-series lung CT dataset, medical image segmentation**

I. Introduction

In recent years, medical imaging research has experienced significant advancements driven by the emergence of deep learning techniques, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs) [1]. These deep learning models have demonstrated remarkable performance in various medical imaging applications, including but not limited to detecting and classifying abnormalities in X-ray, MRI, and CT scans, as well as assisting in disease diagnosis, treatment planning, and prognosis [2], [3]. The importance of medical imaging research lies in its transformative impact on healthcare, enabling faster and more accurate diagnosis, early detection of diseases, personalized treatment strategies, and improved patient outcomes [4].

However, a major problem in the current research is the lack of consideration for the temporal aspect of the data. Many studies focus on individual time points, neglecting the valuable information contained in the sequential nature of the imaging data. This oversight can lead to missed opportunities in understanding disease progression, monitoring changes over time, and providing context for accurate diagnosis [5]. Time-series medical imaging research involves the analysis and interpretation of sequential medical imaging data acquired at different time points for the same patient. Instead of considering isolated images, this approach aims to utilize the temporal information contained within the series of images to gain a more comprehensive understanding of disease progression, treatment response, and overall patient health [6]. The current state of development in time-series medical imaging research has been significantly influenced by the advancements in deep learning techniques, particularly the utilization of RNNs and their variants [7], [8].

Medical imaging data is often collected at irregular intervals, and the number of observations per patient may vary, posing challenges for traditional LSTM networks, which are typically applied on regularly-sampled data [9]. This irregularity can lead to difficulties in applying traditional RNN models like LSTM networks, which are typically designed for regularly-sampled data. Analyzing longitudinal medical imaging data requires effective sequential modeling to capture temporal dependencies and changes over time. To address these limitations, researchers have started exploring novel architectures and modifications to RNNs to handle irregularly-sampled data and capture global temporal variations. One promising approach is the Distanced LSTM (DLSTM) proposed in [10]. The DLSTM incorporates a Temporal Emphasis Model (TEM) to capture global time distance, enabling the network to better understand the timing of different scans and their relevance for diagnosis.

Inspired by the temporal models tLSTM [11] and DLSTM [10], we propose an innovative temporal medical imaging network, which is based on another excellent time series model Transformer [12]. We design a spatial-temporal fusion medical

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image segmentation network to comprehensively analyze lung images by both longitudinal and cross-sectional directions, and to fuse the two types of information. The proposed novel temporal analysis network DTransformer, not only establishes the connection between any two elements in a sequence, but also introduces the distance information between the elements, improves the processing capability of unequal spacing sequences, proposes a better solution to the irregular sampling problem of clinical imaging, and can be generalized to other fields. The following are the contributions of the research we have done in this paper:

- In this paper, we propose a new time series analysis model, DTransformer, which is capable of handling time series information with inconsistent time distance and which we believe can be generalized to other areas.
- We produce a time-series lung image dataset, processed the raw data and labeled it with a refined segmentation mask and the severity of the lung corresponding to each time point, which can be used for segmentation as well as classification tasks.
- We design a novel medical image segmentation network incorporating a temporal network that can fully utilize the image information as well as the temporal information to analyze medical images, and verify the results on our dataset.

II. METHOD

In this section, we describe the proposed model in detail. First of all, our segmentation network is improved based on the 3D UNet network, where the encoder-decoder structure performs feature extraction on the input image [13]–[16]. Secondly, the features of the images of this patient at different time points are learned using an additional temporal network to obtain temporal and spatial information, and then together with the features extracted at the decoder at that time point, they are passed to the decoder in the 3D UNet network to obtain the final segmentation results.

A. 3D UNet Network for 3D Medical Image Segmentation

The segmentation network used in the proposed model is a 3D UNet network, the structure of which is shown in Figure 1. It is a fully convolutional neural network based on an encoder-decoder structure, which differs from the original UNet in that 3D convolution as well as pooling is used instead of the original 2D operation, so that 3D images can be processed. There are four layers of encoder and decoder, and each layer of the decoder receives the incoming features from the encoder of that layer through skip connections, preserving the low-level feature information, which is very important for medical images. The encoder extracts potential features from the input image and learns the feature information using two consecutive 3×3 convolutions. To prevent the network size from increasing as the number of channels is multiplied, the feature map size is compressed using 2×2 maximum pooling downsampling to the next layer. Corresponding to the encoder structure, the decoder also has a four-layer structure. The features up-sampled with the decoder are merged along the channel direction through a skip connection, the up-sampling operation is realized by a 2×2 transpose convolution, followed by two consecutive 3×3 convolutions to adjust the feature dimensions, and after repeating the above operation for four times the up-sampled feature maps are kept at the same size as the input image.

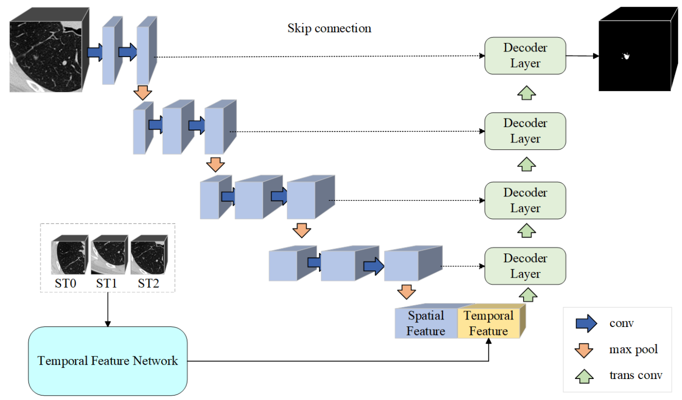

Fig. 1. 3D UNet structure. This network partially fuses temporal features from the temporal network at the bottleneck layer, thus realizing the combination of temporal and spatial information.

We make a change between the encoder and decoder, fusing the image feature information output from the encoder with the temporal information learned by the temporal network, instead of the original convolution operation, so as to realize the fusion of spatial and temporal information, and the subsequent analysis through the decoder. We introduce the structure of the temporal network in detail in the next subsection.

B. The DTransformer Network for Extracting Time-series Image Information

Although many current studies use computers to automatically segment lung nodules, these methods focus on analyzing CT datasets of lung cancer patients at a single point in time, and a great deal of temporal information is ignored. In addition, when physicians analyze these images, they often combine them with CT images taken at other time points of the patient to improve diagnostic accuracy. Therefore, we designed a network that can integrate the image information of patients at different time points, which is used to provide important references and improve the accuracy of the segmentation network. Its structure is shown in Figure 2, which can be viewed as three parts according to the function: detection network, feature extraction network, and temporal feature extraction network.

The detection network is implemented by 3D RPN [17]. The network is a 3D version of the RPN, using a modified 3D UNet [14] as the backbone model. Similar to the detection networks YOLO [19] and SSD [20]. Each time the CT scans are analyzed by 3D RPN and the five highest risk regions (possible nodules) are selected. Its output is directly used as a detection result without the need for additional classifiers.

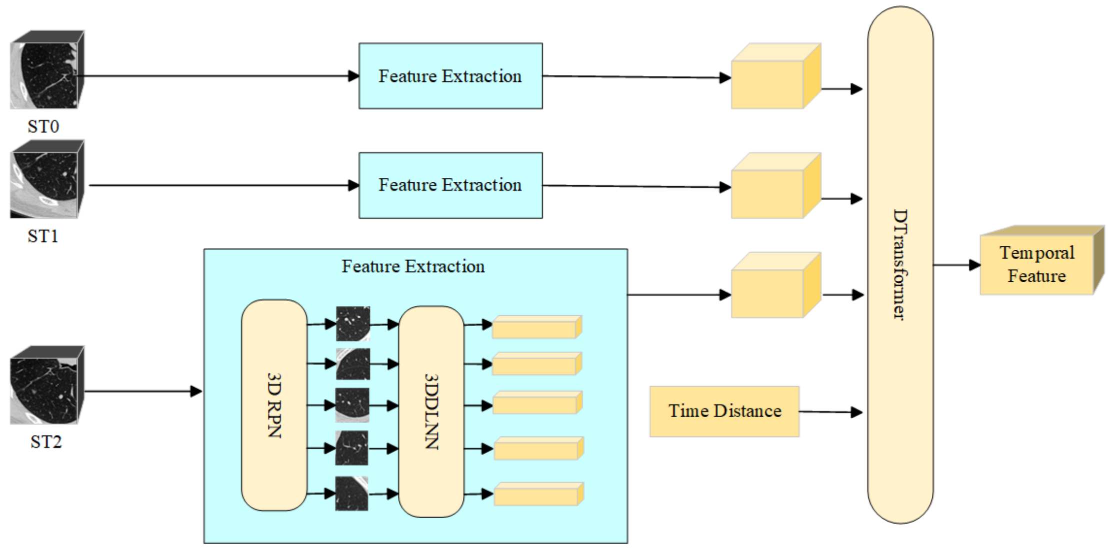

Fig. 2. Time-series image feature extraction network. The images at each time point are subjected to nodule detection by the 3D RPN network [17] and suspicious regions are selected and this feature information is extracted using the 3DDLNN network [18]. The temporal information consisting of features at different time points is passed into the DTransformer to obtain temporal features.

After passing through the detection network, the five highest risk regions are passed into the 3DDLNN model [18], where each region is modeled as a 1D feature. These features are channel-level fused to become 5×64 features. Both parts of the above networks are pre-trained. If a patient has a total of t images from different time points, and each time point image generates a set of features through the detection network as well as the feature network, then these images of the patient generate $t \times 5 \times 64$ features after the above operation. These features are passed into the temporal feature network for subsequent information extraction.

The temporal feature extraction network we designed takes images from all time points of the same patient as independent inputs and is used to analyze the temporal information between the features of the first two networks. Considering that the time elapsed between several consecutive hospital visits is one of the most important bases for diagnosis, a structure that takes into account unfixed time is needed to process temporal data. The previously proposed tLSTM [11] as well as DLSTM [10] is a new LSTM architecture, and no study is found to use the Transformer architecture. Considering the advantages of Transformer itself over LSTM, we have improved the Transformer structure and the altered structure is shown in Figure 3.

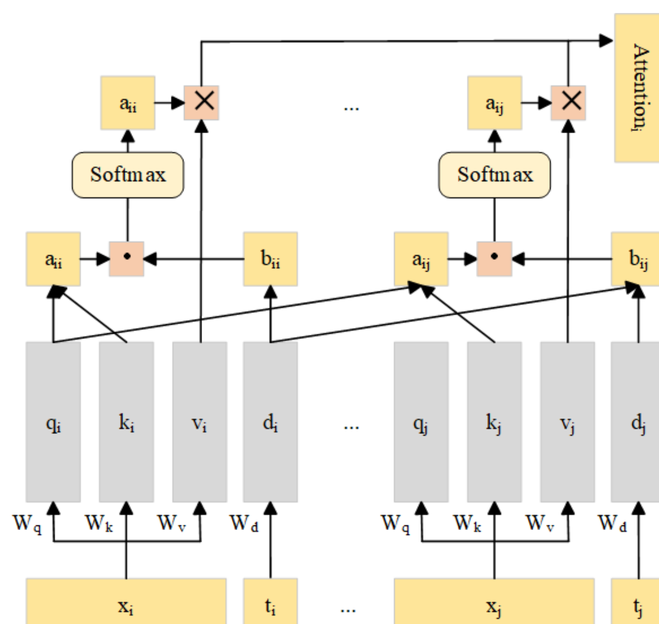

Fig. 3. DTransformer self-attention unit. Distance information is passed into the DTransformer as an additional input, and the distance gap between two elements affect the correlation between the two, and thus the final attention representation.

Self-attention is a key component in the Transformer architecture, which is a popular model for natural language processing (NLP) tasks. It allows the model to weigh the importance of different words or tokens in a sequence while processing them, capturing long-range dependencies effectively. Self-attention enables the Transformer to process sequences in parallel, making it highly scalable and efficient, leading to its significance in achieving state-of-the-art performance in various NLP tasks. However, in previous studies, it was found that the LSTM architecture implicitly assumes that the running time is uniformly distributed among the sequence elements [6]. Similarly we believe that this a priori knowledge still exists in the Transformer architecture. This mechanism will affect the performance of the network when dealing with long sequences. Although it is possible to establish a link between two elements of a sequence through self-attention, this link is cross-sectional, corresponding to the spatial link in an image sequence, and lacks a longitudinal link, i.e., a temporal (or otherwise positional) link. With the introduction of the concept of "distance", DTransformer realizes the creation of information about the distance between two elements in a sequence according to their location (temporal or otherwise), thus incorporating longitudinal information while maintaining the original cross-sectional information.

Specifically, we introduce the distance parameter matrix W^D on its encoder and decoder self-attention computation. Corresponding to the vector d added by DLSTM on top of the vectors h and c of LSTM, the proposed model adds a distance parameter d to the Transformer's attention operation for any two positions in the sequence, compared to the original vector q, k and v. In the computation, the attribute of the ith element in the sequence, in addition to query q_i and key k_i, introduces a vector of distance d_i, that represents its distance information (time distance information in this work), which is computed as $d_i = W^D t_i$, where t_i represents the known distance information corresponding to that element. Passed into the network as an additional additional input, which can be a single or multiple number representation of the vector. In this study, we convert the date on which the patient had a CT taken to a relative number of days, so the input t is a number representing a relative number of days. The self-attention formula in DTransformer is as follows:

$$Attention(Q, K, V, D) = Softmax(\frac{QK^T \cdot DD^T}{\sqrt{dim_k}})V \quad (1)$$

where Q, K, and V are the parameters of the original Transformer, and D is the distance information of all sequences of size $dim_k \times N$, dim_k is the feature dimension of K, N is the length of the sequences, and the size of D is the same as Q and K.

III. Experiment

In this section, we introduce the dataset we use as well as the exploration process, and the preprocessing process. Comparative experiments are performed on several traditional CNNs as well as a recently proposed classical temporal model, and the effectiveness of the proposed temporal model is verified.

A. Dataset

We collected time-series CT data from 99 lung cancer patients at the hospital, with 3 time points of CT for each patient. The source of our dataset is from the First Affiliated Hospital of Dalian Medical University. The data of our study is collected with the informed consent of all participants and in accordance with the ethical guidelines of the Ethics Committee of the Hospital. Clinicians at the hospital collected the raw CTs, labeled the location of the lung nodules at each time point for each lung cancer patient, and then tracked the lung nodules for all lung cancer patients and labeled the same lung nodules in the CTs at each time point for each patient. We generate a time-series CT dataset of lung cancer patients by cropping the region around the lung nodules, including the preprocessed images as well as their exact segmentation masks. The detailed processing is described in our previous work [21].

In our previous work [21], we explored the segmentation effect of time-series images under different time-point inputs using the time-series model. The best and most accurate prediction performance of the model is achieved when multiple time points of CT of lung nodules are used as inputs, which proves the validity and usability of our dataset. This shows that it is feasible to use time-series networks to improve segmentation accuracy.

B. Data Preprocessing

For the spatial network input, the original CTs of lung cancer patients are resampled using a linear interpolation algorithm to make the scanning distance consistent. The lung region is highlighted by adjusting the window width and window position to enable the model to learn diseased as well as healthy regions in the lungs and reduce the influence of other organs and tissues. For training, we divide the image data of all patients at all time points, with 80% of the patient data as the training set and the remaining 20% of the patient data as the test set. Three time-point images are selected for each patient, and the image size of a single time-point is $32 \times 128 \times 128$, so the three time-point images are spliced to be $96 \times 128 \times 128$, which is used as the input of the segmentation network for extracting spatial information.

For the temporal feature network, the images at each of our time points follow the preprocessing in [18], allowing the images to be better adapted to the pre-trained 3D RPN model. A masking operation is performed on other tissues in the original CT image, keeping only the lungs, as some of them may be spherical in shape, which can cause the detection network to misjudge these as nodules. By extracting the lung mask, only the lung region is preserved to exclude interference in the detection phase. Afterwards, some tissues as well as nodules in the lungs are highlighted by window-tuning operations. After the above pre-processing, the original image is processed into a $128 \times 128 \times 128$ size image, which is passed into the 3D RPN in the temporal feature network.

C. Implementation Details

The CUDA version of the experimental platform is 12.0, and the graphics card is NVIDIA GeForce RTX 4090. The batch size is set to 3 during the training process, and Adam is used as the optimizer and the initial learning rate is set to 0.0001, which decays as the number of epochs increases. Considering that the lesion region in the image only accounts for a small part of the whole image, the loss function used is the Dice loss commonly used in medical image segmentation, which is used to solve the problem that the cross-entropy loss fails when facing the imbalance of positive and negative samples.

In addition, in order to evaluate the effect of the segmentation model in a comprehensive and integrated manner, we selected a variety of evaluation metrics commonly used in

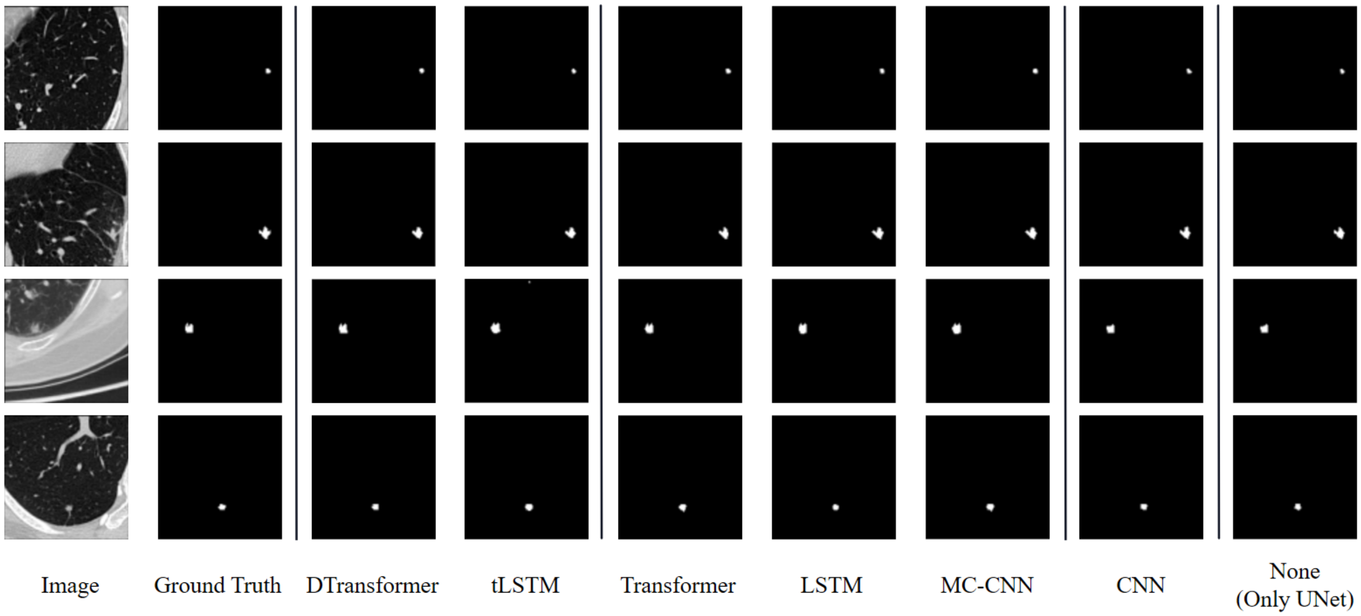

Fig. 4. 3D UNet introduces prediction results for different temporal models.

medical segmentation to assess the model effect. These metrics are Dice score, sensitivity, and Hausdorff distance.

D. Results

In order to validate the segmentation performance of 3D UNet networks after adding different temporal models, we conducted experiments on our dataset using a single 3D UNet as well as models fusing various types of temporal networks, and compared them on the metrics introduced in the previous section via Table I. The horizontal lines in the table delineate how much information these models obtain from the original input, with the amount of information increasing from top to bottom. The information added is: the original image (spatial information), features at a single point in time, features at multiple points in time (temporal information), and the time distance between different points in time (temporal information). Where the CNN network only inputs the image from the last time point, the MC-CNN network is inputting the images from multiple time points as different channels. Despite the addition of the temporal network compared to a traditional segmentation network, this is negligible to the network size of the image as these temporal features are computed by the pre-trained network and are much smaller in scale compared to the 3D image, being linear. It is also almost difficulty-free in terms of information acquisition difficulty, as the time of capture of the image is saved along with the image.

By comparing the single 3D UNet network with other models fused with temporal sequences, we find that the segmentation of the network is improved to varying degrees with the addition of temporal information, and the improvement is significant, suggesting that this spatial-temporal feature fusion method is very effective. In addition, in the comparison of

TABLE I
SEGMENTATION PERFORMANCE COMPARISON OF 3D UNET NETWORKS AFTER FUSING DIFFERENT TEMPORAL MODELS

Method	Dice(%)	Sensitive(%)	H95
3D UNet [14]	62.58	54.32	14.66
3D UNet+CNN [18]	65.90	57.82	**9.90**
3D UNet+MC-CNN [10]	67.78	63.21	20.79
3D UNet+LSTM [22], [23]	68.84	59.66	14.17
3D UNet+Transformer [12]	<u>69.35</u>	**65.37**	24.08
3D UNet+tLSTM [11]	66.68	62.01	17.23
3D UNet+DTransformer	**70.04**	<u>64.09</u>	<u>13.34</u>

models with added temporal networks, DTransformer outperforms other models by a slight advantage, proving that the new temporal network proposed in this paper has better ability to extract temporal information and is well adapted to our dataset.

Figure 4 illustrates the segmentation prediction results produced by the 3D UNet segmentation model when different temporal models are introduced. Although the original 3D UNet as well as its combination with each of the temporal models achieves the expected segmentation results, there are still subtle differences. Although each temporal model achieves a shape consistent with Ground Truth, DTransformer outperforms the other temporal models in terms of detailing. It is sufficient to accomplish the segmentation task by spatial information alone, as confirmed by many studies as well as excellent models. However, if the temporal features are combined, the model will have more detailed segmentation results, which are more favorable for diagnosis. From Table I, it can be seen that the introduction of more information is beneficial to the enhancement of the network, but more important is how to make full use of this information, and it is necessary to design the network structure for the characteristics of the time-series information. The bolded portion of each column in the table is the optimal result and the underlined portion is the sub-optimal result.

IV. Conclusion

In this paper, we propose a cross-sectional-longitudinal analysis network based on spatial and temporal fusion and design a novel temporal network, DTransformer, for realizing the processing of unequally spaced sequences, which is validated on our temporal lung CT dataset. The proposed network contains the spatial network 3D UNet with the temporal network DTransformer, which enables cross-sectional and longitudinal analysis. After performing comparative analysis, it is shown on our dataset that this fusion approach is effective and the proposed temporal model, DTransformer, achieved excellent performance. DTransformer may be advantageous in subsequent studies of unequally spaced sequences, generalizing to other temporal analysis tasks or other cases of unequally spaced factors. In the future, we will continue to refine our dataset, validate the performance of DTransformer on more tasks, and improve its network structure.

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